

# Estimating a Population Mean with the Central Limit Theorem

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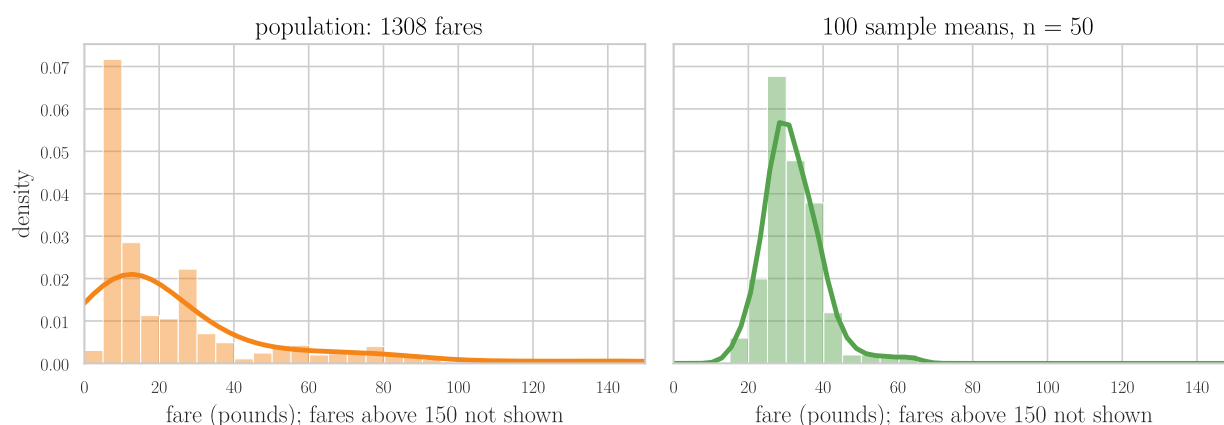
Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Normal distribution ([Note 250](#)); Sampling distribution and standard error ([Note 271](#)).
- **Leads to:** Confidence intervals ([Note 280](#)); Z-test and rejection regions ([Note 291](#)).

## 1. Overview

**Key point:** We pretend not to know the average Titanic fare, estimate it from random samples with the central limit theorem, and check the estimate against the truth.



**Figure 1:** Titanic fares: the population is right-skewed (left), the means of 100 samples of 50 passengers are close to a bell (right)

The [central limit theorem Note](#) showed that sample means are approximately normal, centred on the population mean  $\mu$ , with standard error  $\sigma/\sqrt{n}$ . This Note uses that result on real data.

The Titanic carried 1309 passengers, and we know every fare. We act as if we did not, estimate the average fare from samples, and only at the end compare with the true value. Figure 1 shows why this is a fair test: the fares are far from normal, yet the sample means form a bell.

The Note covers:

- turning the sample means into a **point estimate** of  $\mu$ ;
- turning the standard error into a **range** that probably contains  $\mu$ ;
- a mistake in the divisor of the standard error, and the right one;
- the same steps as a template for estimating a country's average income.

## 2. The population: every Titanic passenger

**Key point:** The Kaggle train file (891 passengers) and test file (418 passengers) together hold all 1309 passengers; their 1308 known fares are our population.

The Titanic data comes in two files. The train file has 891 passengers with a **Survived** column; the test file has the other 418 passengers without it. For the fares, the split does not matter: stacking both files gives every passenger on board, the whole population.

**Python:** Building the population.

```
import pandas as pd
```

```
train = pd.read_csv("data/titanic_train.csv")
test = pd.read_csv("data/titanic_test.csv")
df = pd.concat([train.drop(columns="Survived"), test])
df = df.sample(frac=1, random_state=42)  # shuffle the rows
fare = df["Fare"].dropna()              # 1308 known fares
pd.concat stacks the two tables. The Survived column is dropped first, because the
test file has none. One fare is missing, so dropna() leaves 1308.
```

The fares are strongly right-skewed (Figure 1, left): skewness 4.37, median 14.45 pounds, maximum 512.33 pounds. Most passengers paid little; a few first-class passengers paid a great deal. No one would call this distribution normal.

## 3. A sampling distribution of the mean fare

**Key point:** 100 random samples of 50 passengers give 100 sample means, the sampling distribution of the mean fare.

We now draw samples as if surveying passengers:

1. Draw 50 passengers at random (sample size  $n = 50$ , above the usual threshold of 30).
2. Record their fares.
3. Repeat until we have 100 samples.
4. Compute the mean of each sample.

The first five sample means are 37.27, 25.21, 34.21, 28.85 and 27.57 pounds: every sample gives a different answer. Figure 1 (right) shows all 100. They are much closer to a bell than the fares, as the CLT predicts.

The bell is not perfect: the 100 means still have a skewness of 1.12. For a population as skewed as these fares,  $n = 50$  gives a shape that is roughly normal, not exactly normal.

**Python:** 100 samples of 50.

```
import numpy as np
```

```
rng = np.random.default_rng(42)  # fixed seed
```

```

samples = np.array([fare.sample(50, random_state=rng).to_numpy()
                    for _ in range(100)])
sample_means = samples.mean(axis=1)  # one mean per sample

```

samples has shape (100, 50): 100 samples, 50 fares each. Within one sample, fare.sample(50) never picks the same passenger twice.

## 4. The point estimate

**Key point:** The average of the sample means, 31.87 pounds, is our single best guess for the population mean: a point estimate.

By the CLT, the sample means are centred on the population mean. So their average is our best guess for  $\mu$ . A single number used as the best guess for an unknown parameter is a **point estimate**.

1. **In words:** average the  $k$  sample means.
2. **Formula:**

$$\hat{\mu} = \bar{\bar{x}} = \frac{1}{k} \sum_{j=1}^k \bar{x}_j$$

The hat on  $\hat{\mu}$  marks an estimate of  $\mu$ ;  $\bar{\bar{x}}$  (“x double bar”) is the mean of the means.

3. **Example:** for our 100 sample means,

$$\hat{\mu} = \frac{37.27 + 25.21 + 34.21 + \dots}{100} = 31.87 \text{ pounds}$$

A point estimate is almost never exactly right. The CLT says it is close to  $\mu$ , not equal to it. So instead of claiming “the average fare is 31.87 pounds”, we give a range, just as a weather forecast says “28 to 32 degrees tomorrow” rather than one exact temperature.

## 5. A range for the population mean

**Key point:** About 95% of a normal distribution lies within 2 standard deviations of its centre, so “estimate  $\pm$  2 standard errors” gives a range that contains  $\mu$  about 95% of the time.

The point estimate is itself a mean of normal-shaped values, so it follows a normal distribution around  $\mu$ . By the 68-95-99.7 rule (see the [standard normal Note](#)), it lands within 2 of its standard deviations of  $\mu$  about 95% of the time. Turned around: the range “estimate  $\pm$  2 standard deviations” contains  $\mu$  about 95% of the time.

The standard deviation of an estimate is its standard error (SE). Our estimate averages 100 independent sample means, each with spread  $s_{\bar{x}}$  (the standard deviation of the 100 sample means). By the CLT applied once more, averaging 100 of them divides that spread by  $\sqrt{100}$ .

1. **In words:** the standard error of the average of  $k$  sample means is the standard deviation of the sample means divided by the square root of  $k$ , the number of samples.

## 2. Formula:

$$SE = \frac{s_{\bar{x}}}{\sqrt{k}}, \quad \text{range} = \hat{\mu} \pm 2 \times SE$$

3. **Example:** the 100 sample means have standard deviation  $s_{\bar{x}} = 7.56$  pounds, so

$$SE = \frac{7.56}{\sqrt{100}} = \frac{7.56}{10} = 0.756, \quad 31.87 \pm 2 \times 0.756 = 31.87 \pm 1.51$$

The range is **30.35 to 33.38 pounds**.

Now we look at the truth: the mean fare of all 1308 passengers is **33.30 pounds**. The true mean lies inside the range, near its upper end. The point estimate alone (31.87) was off by 1.43 pounds; the range says honestly how far off it might be.

The 7.56 itself confirms the CLT: it should be  $\sigma/\sqrt{50}$ , and the population standard deviation of the fares is  $\sigma = 51.74$ , giving  $51.74/\sqrt{50} = 7.32$ .

**Extra:** The value 2 is a rounded number. The exact multiplier that leaves 95% in the middle of a normal curve is 1.96, from the z-table:  $\Phi(1.96) = 0.975$  (see the [standard normal Note](#)). With 1.96 the range here becomes  $31.87 \pm 1.48$ . The [confidence intervals Note](#) derives it.

## 5.1 Why 2 standard errors

**Key point:** A wider range is more likely to contain  $\mu$  but says less; 2 standard errors (95%) is the usual compromise.

We could use 1, 2 or 3 standard errors:

| Range      | Share of a normal curve | Fare range (pounds) |
|------------|-------------------------|---------------------|
| $\pm 1$ SE | 68%                     | 31.11 to 32.62      |
| $\pm 2$ SE | 95%                     | 30.35 to 33.38      |
| $\pm 3$ SE | 99.7%                   | 29.60 to 34.13      |

With 1 SE, the range is narrow but wrong about a third of the time; here it misses 33.30. With 3 SE, we are almost never wrong, but the range is wider.

A range must be narrow enough to be useful. “The average salary in a company is between 10,000 and 15,000 rupees, and I am 95% sure” says more than “between 10,000 and 45,000 rupees, and I am 99% sure”. The 95% level, about 2 standard errors, is the standard compromise.

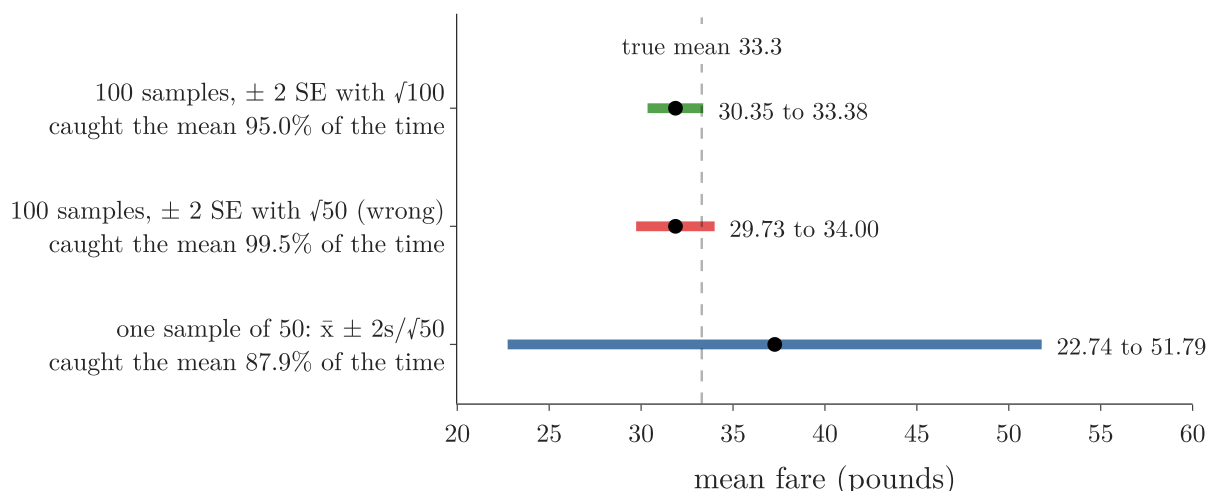
## 6. A common mistake: dividing by the wrong square root

**Key point:** The standard error of an average of 100 sample means divides by  $\sqrt{100}$ , the number of samples; dividing by  $\sqrt{50}$ , the sample size, mixes up the two sizes.

Two sizes appear in this method, and they are easy to swap:

- $n = 50$ , the **sample size**: the standard deviation of the sample means is already  $\sigma/\sqrt{50}$ ; the division by  $\sqrt{n}$  has happened once, inside the CLT.
- $k = 100$ , the **number of samples**: averaging the 100 sample means divides their spread by  $\sqrt{100}$ .

Dividing  $s_{\bar{x}} = 7.56$  by  $\sqrt{50}$  instead gives  $7.56/7.07 = 1.07$  and the range 29.73 to 34.00. This wrong range also contains 33.30, but it is the wrong width. Figure 2 shows what each version does when the whole experiment is repeated 1000 times.



**Figure 2:** Three ranges for the mean fare against the true mean 33.30; each label gives how often that method caught the true mean in 1000 repetitions

- $\sqrt{100}$  (**right**): the range contains the true mean in 95.0% of the repetitions, exactly as promised.
- $\sqrt{50}$  (**wrong**): 99.5%. The range is about 40% too wide, so the “95%” label is false.

## 7. With only one sample

**Key point:** In practice we have one sample, and the range is  $\bar{x} \pm 2s/\sqrt{n}$ ; for very skewed data,  $n = 50$  is too small for this to be reliable.

Drawing 100 samples is a teaching device: it lets us see the sampling distribution. A real survey collects one sample. Then the CLT is used directly: the one sample mean  $\bar{x}$  is the point estimate, and the standard error  $\sigma/\sqrt{n}$  is estimated with the sample’s own standard

deviation  $s$ :

$$\bar{x} \pm 2 \frac{s}{\sqrt{n}}$$

For the first of our samples,  $\bar{x} = 37.27$ ,  $s = 51.34$ , so the range is  $37.27 \pm 2 \times 51.34/\sqrt{50} = 37.27 \pm 14.52$ : **22.74 to 51.79 pounds** (Figure 2, blue). The one-sample range contains 33.30, and it is much wider: its standard error is  $\sigma/\sqrt{50}$ , while the average of 100 samples of 50 has standard error  $(\sigma/\sqrt{50})/\sqrt{100} = \sigma/\sqrt{5000}$ , ten times smaller.

By the same formula, 100 samples of 50 give the same standard error,  $\sigma/\sqrt{5000}$ , as one large sample of 5000. If we have several samples, we can pool them into one.

**Extra:** The one-sample range with  $n = 50$  caught the true mean in only 87.9% of the 1000 repetitions of Figure 2, not 95%. The fares are so skewed that only 38 of the 1308 passengers paid more than 200 pounds. The Notebook tests whether missing those tickets is the cause, with 2000 fresh samples of 50. Of their 221 misses, 218 fell below the true mean. A sample with no fare above 200 missed 44.6% of the time; a sample with at least one missed 1.2% of the time. The sample mean and  $s$  move together (correlation 0.86), so a sample without the expensive tickets has both  $\bar{x}$  and  $s$  too small at once: the range is too low and too narrow. With  $n = 200$  the share rises to 95.6%. The  $n \geq 30$  rule is not enough for extremely skewed data. Using  $s$  in place of  $\sigma$  also adds its own uncertainty; the [t-procedure Note](#) handles that part.

## 8. Template: the average income of a country

**Key point:** Random representative samples of at least 30, their means, the average of the means, its standard error, and a 95% range around it.

The same steps estimate any population mean, such as the average yearly income in India:

1. **Collect several random samples** of incomes, each with  $n \geq 30$ . They must be representative: not only from cities, not only from one state, not only from one age group.
2. **Compute the mean of each sample**, which gives the sampling distribution of the mean.
3. **Average the sample means**: this is the point estimate of the population mean.
4. **Compute the standard error**: the standard deviation of the sample means divided by the square root of the number of samples.
5. **Build the 95% range**: point estimate  $\pm 1.96 \times \text{SE}$ .
6. **Report** the estimate together with the range.

We have no real income data, so the Notebook simulates a log-normal population of one million yearly incomes (see the [uniform and log-normal Note](#)). From 100 samples of 50 people, the estimate is 1,28,148 rupees with 95% range 1,25,386 to 1,30,911 rupees; the true mean of the simulated population is 1,27,886 rupees.

The result is only as good as the samples. Biased samples (see sampling bias in the [challenges in ML Note](#)) give a tight, confident range around the wrong value.

## 9. Summary

| Step                           | Formula                                                | Titanic fares                |
|--------------------------------|--------------------------------------------------------|------------------------------|
| Sample means                   | $\bar{x}_1, \dots, \bar{x}_{100}$ , each from $n = 50$ | 37.27, 25.21, 34.21, ...     |
| Point estimate                 | $\hat{\mu}$ = average of the sample means              | 31.87                        |
| Spread of the sample means     | $s_{\bar{x}} \approx \sigma/\sqrt{n}$                  | 7.56 (CLT: 7.32)             |
| Standard error of the estimate | $s_{\bar{x}}/\sqrt{k}$                                 | $7.56/\sqrt{100} = 0.756$    |
| 95% range                      | $\hat{\mu} \pm 2 SE$                                   | 30.35 to 33.38 (truth 33.30) |
| One sample only                | $\bar{x} \pm 2s/\sqrt{n}$                              | 22.74 to 51.79               |

- The CLT works on the skewed fares: the sample means are close to a bell.
- A range says how far off the point estimate might be; 2 standard errors is the 95% compromise.
- Divide by the square root of the number of samples when averaging sample means; the sample size is already inside  $s_{\bar{x}}$ .
- For extremely skewed data, a sample of 50 can be too small; and no sample size fixes a biased sample.

## 10. Key terms

| Term                      | Meaning                                                                                         |
|---------------------------|-------------------------------------------------------------------------------------------------|
| Point estimate            | A single number computed from sample data as the best guess for an unknown population parameter |
| $\hat{\mu}$               | An estimate of the population mean $\mu$                                                        |
| Number of samples ( $k$ ) | How many samples are drawn; different from the sample size $n$                                  |